SYSDESIGN REFERENCE

Database Indexing Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Trade write and storage cost for faster access paths aligned to real query predicates.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / DATABASE-INDEXING

SYSDESIGN

Database Indexing Tradeoff Matrix

CORE IDEA

Trade write and storage cost for faster access paths aligned to real query predicates.

Reference architecture decision chain

01	02	03	04
Composite Order	Selectivity	Covering Indexes	Range Scans

Decisions to defend

+ Composite order
+ selectivity
+ covering indexes
+ range scans
+ and maintenance.

Failure modes to test

+ Unused indexes
+ write amplification
+ low-selectivity scans
+ and query-plan drift.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Database Indexing Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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